ModelIt!

Student Activity Pack

Why Can't You See in the Dark?

The Surprising Science of Light and Vision

Grade 4 | NGSS 4-PS4-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think you cannot see in a completely dark room?

2 Do your eyes send something OUT to see things, or do they take something IN?

3 Draw a picture showing how you think you see a book on a table.

4 What is the difference between something that makes light and something that just reflects light?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Light Source Brightness | Light Bouncing Off Objects | Eye Detection

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Pitch Black Room

Light Source Brightness set to zero

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Bright Sunny Day

Light Source Brightness set to maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Dim Flashlight

Light Source Brightness set to 20%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Object Color — Different colors reflect different amounts of light — white reflects the most, black absorbs the most

Object Distance — How far away an object is — light spreads out over distance, making far-away objects dimmer

Pupil Size — Your pupil gets bigger in the dark to let in more light and smaller in bright light to protect your eye

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If your eyes are open in a completely dark room, why can you not see anything at all? Use evidence from your simulation to support your answers.

1 Why can you see the Moon at night even though it does not make its own light?

2 If light did not bounce off objects, what would the world look like?

3 Why is it easier to see a white shirt in the dark than a black shirt?

4 How are your eyes similar to a camera?

5 How did your model help you understand why darkness means invisibility?

8 STEM Challenge: Design a Light Maze

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YOUR ENGINEERING MISSION

Design a maze where light must bounce off mirrors to reach a target, proving that we see objects because light reflects off them.

THE SCENARIO

A science museum wants an interactive exhibit that teaches visitors how light and vision work. Your engineering team must design a light maze where visitors use mirrors to bounce a flashlight beam through a maze to hit a target. The exhibit must prove that seeing requires light to bounce off objects.

Engineering Design Thinking

1. HOW MANY MIRRORS WILL YOU NEED AND WHERE WILL YOU PLACE THEM?
2. WHAT ANGLE DO THE MIRRORS NEED TO BE AT TO BOUNCE LIGHT IN THE RIGHT DIRECTION?
3. WHAT MATERIAL WILL YOUR TARGET BE MADE OF SO IT IS EASY TO SEE WHEN LIGHT HITS IT?
4. HOW WILL YOU BLOCK LIGHT FROM TAKING SHORTCUTS THROUGH THE MAZE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Optical Engineers and Vision Scientists study how light behaves and design things like cameras, telescopes, eyeglasses, and laser systems. They earn $70,000–$120,000/year.